Lesson 17: Non-Experimental and Correlational Research

English Stage 4 (Psychology) · Unit 3: Research Designs

Adapted from Research Methods in Psychology (4th American edition) by Rajiv S. Jhangiani, I-Chant A. Chiang, Carrie Cuttler and Dana C. Leighton (CC BY-NC-SA 4.0): 6.1: Prelude to Nonexperimental Research; 6.2: Overview of Non-Experimental Research; 6.3: Correlational Research.

In this lesson

You will learn when researchers measure variables instead of manipulating them, the main types of non-experimental research, how to read a correlation (Pearson's *r*), and why correlation does not imply causation. You will practise the difference between causal language and the language of association.

Learning goals

After this lesson you should be able to:

- Define non-experimental and correlational research and explain when each is preferred.
- Distinguish cross-sectional, longitudinal and cross-sequential studies.
- Interpret the direction and strength of Pearson's *r*, and recognise nonlinear relationships and restriction of range.
- Explain the directionality and third-variable problems.
- Report relationships without claiming causation.

Key language

Term or phrase	Meaning	Example
non-experimental research	research that lacks the manipulation of an independent variable; variables are measured as they naturally occur	Milgram's original obedience study
correlational research	measuring two variables, with little or no control of extraneous variables, and assessing the relationship between them	self-esteem and GPA
cross-sectional / longitudinal study	comparing pre-existing groups (e.g., ages) / following one group over time	A difference between age groups may be a cohort effect.
positive / negative relationship	higher scores on one variable go with higher / lower scores on the other	stress and immune system functioning (negative)
Pearson's *r*	a measure of the strength of the correlation between two quantitative variables, from −1.00 to +1.00	$r = +.51$

Term or phrase	Meaning	Example
nonlinear relationship	as X increases, Y does not change at a constant rate; best described by a curve	sleep and depression
restriction of range	a limited range of scores on X or Y in the sample, which makes a relationship look weaker	only 18- to 24-year-olds
directionality problem	X and Y are related, but Y may cause X instead of X causing Y	Does happiness cause exercise?
third-variable problem	a third variable, Z, may cause both X and Y	physical health
is statistically related to / is associated with	language of association, not causation	Higher stress is associated with lower immune system functioning.
it stands to reason that	it is logical that	It stands to reason that non-experimental research is appropriate when these conditions are not met.

Rules and explanations

The ideas

Non-experimental research (Sections 6.1-6.2)

The four classic studies in Section 6.1 (Milgram, Loftus and Pickrell, Cacioppo and Petty, Rosenhan) are not experiments. Non-experimental research usually cannot show causation, but it is not less important. It is preferred when:

1. the question is about a **single variable** (How accurate are first impressions?);
2. the relationship is **not thought to be causal** (verbal and mathematical intelligence);
3. the question is causal, but the IV **cannot be manipulated** or participants cannot be randomly assigned (hippocampus damage and long-term memory);
4. the question is **broad and exploratory**, or about an experience (being a working mother diagnosed with depression).

To **explain** causes, experiments are preferred; to **describe** or **predict**, non-experimental research is appropriate. The approaches can work together: Milgram's original study measured one variable (obedience) under the same conditions for everyone; his later experiments manipulated, for example, the distance between participant and confederate.

Types. In **correlational research**, two variables are measured and their relationship assessed. **Observational research** observes behaviour without manipulating anything (in Loftus and Pickrell's study, nearly a third of participants "remembered" at least one event that never happened). Studies of change over time:

Design	How	Strength / limitation
Cross-sectional	compare age groups at one time	quick, but differences may be **cohort effects** (due to the generation people come from, not age)
Longitudinal	follow one group as it ages	better for ageing, but very time-consuming
Cross-sequential	follow several age groups for a shorter period (e.g., ages 20-60, for ten years)	immediate comparisons, and it separates age effects from cohort effects

All of these are **quantitative** (numbers analysed statistically); in **qualitative** research the data are usually non-numerical, such as the notes and hospital records in Rosenhan's study.

Internal validity is generally highest in experiments, lower in quasi-experiments and lowest in correlational studies, but the ranges overlap: a poorly designed experiment with many confounds can be lower than a good quasi-experiment.

Correlational research (Section 6.3)

Reasons to choose a correlational study:

- to describe and predict a relationship that is not thought to be causal;
- the IV cannot be manipulated (cannabis use and memory: manipulating cannabis use would be unethical);
- to establish the **reliability and validity** of measures (a brief and a long extraversion test); here *independent* and *dependent variable* do not apply;
- **external validity** is often higher, because nothing is manipulated or controlled;
- **converging evidence** with experiments (violent television and aggression).

What makes a study correlational is **how it is conducted**, not the variables or the statistics. Giving the Rosenberg Self-Esteem Scale to 50 American and 50 Japanese students "feels" like an experiment, but nationality was not manipulated. In the to-do list example, randomly assigning people to make lists makes it an experiment; asking whether they make lists makes it correlational.

Pearson's *r*

- In a **scatterplot**, each point shows one person's scores on both variables (stress and physical symptoms: $r = +.51$).
- r runs from −1.00 to +1.00; 0 means no relationship (a shapeless "cloud" of points).
- Guidelines: about ±.10 small, ±.30 medium, ±.50 large. The sign shows direction, not strength: +.30 and −.30 are equally strong. Most correlations in psychology are small or moderate.
- r can mislead in two situations: **nonlinear relationships** (people who sleep about eight hours are least depressed, so r is close to zero even though the relationship is fairly strong; always make a scatterplot first) and **restriction of range** (age and enjoyment of hip hop: $r = -.77$ overall but 0 for 18- to 24-year-olds).

Correlation does not imply causation

Chocolate consumption and Nobel prizes correlate (r = 0.79), but chocolate does not win prizes. Two reasons:

- **Directionality problem:** exercise may cause happiness, or happiness may cause exercise.
- **Third-variable problem:** European geography may explain both chocolate consumption and investment in education and technology; physical health may cause both exercise and happiness. Such correlations are called **spurious correlations**.

The most effective solution is an experiment: randomly assign people to 15 minutes on a treadmill or on a couch. Then mood cannot have affected how much they exercised, and no third variable can explain the difference.

Language focus

Causal language versus the language of association. A correlational result must be reported with verbs and phrases that describe a relationship:

Association (correlational studies)	Causation (experiments only)
"people under more stress **tend to have** more physical symptoms"	"making the daily to-do lists **reduced** participants' stress"
"higher stress **is associated with** lower immune system functioning"	"Lots of Candy **Could Lead to** Violence"
"these variables **are statistically related**"	"exercising **causes** happiness"

Note *tend to* + base verb, and the fixed prepositions *associated* **with**, *correlated* **with**, *related* **to**, *the relationship* **between** *X and Y*.

> **Note for Arabic speakers** - Prepositions do not translate one to one. Learn each phrase whole: *associated with* (not *associated to*), *correlated with*, *related to*, *a relationship between X and Y*.

Practice

1. Reading questions

1. Give the four situations in which non-experimental research is preferred.
2. How did Milgram use both non-experimental and experimental approaches?
3. What is a cohort effect, and how does a cross-sequential study deal with it?
4. Why is the Rosenberg Self-Esteem Scale study of American and Japanese students correlational?
5. What decides whether the to-do list study is an experiment or a correlational study?
6. What do the sign and the size of Pearson's r tell you? Which is stronger: −.40 or +.30?
7. Why should you make a scatterplot before computing Pearson's r?

2. Book questions and invented data

a. From the box "Lots of Candy Could Lead to Violence": children who ate candy every day were more likely to be arrested for a violent offense later in life. "What alternative explanations can you think of for this statistical relationship? How could the headline be rewritten so that it is not misleading?"

b. Invented data for practice: average nightly sleep (one-week diary) and words recalled (out of 30) for ten students.

Student	1	2	3	4	5	6	7	8	9	10
Sleep (hours)	4.5	5.0	5.5	6.0	6.5	7.0	7.0	7.5	8.0	8.5
Words recalled	12	16	13	15	17	17	19	18	18	16

1. Sketch a scatterplot. Is the relationship positive or negative?
2. For all ten students, $r = +.71$. How strong is this by the book's guidelines?
3. For students 6 to 9 only (7 to 8 hours of sleep), $r = .00$. Which problem does this show, and how would you avoid it?
4. Does $r = +.71$ show that more sleep improves memory? Give one directionality explanation and one third variable.
5. Student 10 slept the most but recalled fewer words than students 6 to 9. If many long sleepers scored lower, what kind of relationship might this be, and why would r mislead?

c. Practice (Section 6.7): "For each of the following statistical relationships, decide whether the directionality problem is present and think of at least one plausible third variable." (1) "People who eat more lobster tend to live longer." (2) "People who exercise more tend to weigh less." (3) "College students who drink more alcohol tend to have poorer grades."

3. Language exercise

A. Does each sentence claim causation (C) or only a relationship (R)?

1. "people who exercise are happier on average than people who do not"
2. "perhaps being stressed has a negative effect on people's ability to plan ahead"
3. "nations that have won more Nobel prizes tend to have higher chocolate consumption"

B. Rewrite as findings of a correlational study.

4. Sleeping more improves students' memory.
5. Eating candy every day leads to violence.

Speak with your tutor

1. In PSYC 110 Lecture 13, Professor Bloom criticises a study suggesting that family meals keep teenagers away from drugs. Which of his criticisms is a directionality problem, and which a third-variable problem?
2. In PSYC 110 Lecture 18, Professor Nolen-Hoeksema shows that 15- to 24-year-olds have the highest rates of depression and mentions that younger generations may be more prone to

depression. Which explanation is a cohort effect? Design a cross-sequential study to test it.

3. In PSYC 110 Lecture 15, Professor Bloom says that in Milgram's research "you get less of an effect if you have to touch the guy". Explain how Milgram could establish this with an experiment after his original non-experimental study.

Writing task

Research project: a correlational alternative (Lessons 17-18; see also Lessons 19-22 for other alternatives).

Write 200-250 words describing a correlational version of your study of sleep and memory in university students. Include:

- the two variables and how each will be measured, not manipulated (for example, a one-week sleep diary and a standard word-recall test);
- the expected direction of the relationship, checked with a scatterplot and Pearson's *r*;
- why a correlational design may be useful (real sleep habits, external validity, ethics);
- how you will avoid restriction of range;
- the directionality and third-variable problems, with one example of each;
- one sentence comparing this design with your experiment (internal versus external validity).

Language: association language (*tend to*, *is associated with*, *is positively correlated with*); no causal verbs in the conclusions.

If you work on your own topic, describe a correlational version of your own study.

Self-editing checklist:

- Both variables are measured; nothing is manipulated.
- The predicted direction (positive or negative) is stated.
- Restriction of range and nonlinearity are considered.
- One directionality example and one third variable are given.
- No sentence claims that sleep *improves* or *causes* anything.

Paste your draft into the chat and ask your tutor for feedback. Then rewrite it.

Watch online

- Interpreting Correlations, an interactive visualization by Kristoffer Magnusson (recommended in the section): https://rpsychologist.com/correlation/
- Spurious Correlations by Tyler Vigen (the source of the section's example): https://tylervigen.com/spurious-correlations
- PSYC 110 Lecture 13 (the family-meals study and correlation versus causation): https://oyc.yale.edu/psychology/psyc-110/lecture-13

Answer key

1. Reading questions

1. A single-variable question; a non-causal relationship; a causal question in which the IV cannot be manipulated or participants cannot be randomly assigned; a broad, exploratory question or one about an experience (6.2, "When to Use Non-Experimental Research").
2. His original study measured one variable, obedience, with all participants under the same conditions; later experiments manipulated IVs such as distance and location (same subsection).
3. Differences between age groups caused by the generation people come from rather than by age. A cross-sequential study follows several age groups for a period, so it can tell whether the original differences are age effects or cohort effects (6.2, "Cross-Sectional, Longitudinal, and Cross-Sequential Studies").
4. The researcher did not manipulate nationality; the variables were measured (6.3, box "Does Correlational Research Always Involve Quantitative Variables?").
5. How it was conducted: random assignment to making lists or not makes it an experiment; simply asking makes it correlational (same box).
6. The sign gives the direction; the absolute value gives the strength. −.40 is stronger (6.3, Pearson's *r* paragraph).
7. *r* is a good measure only for linear relationships; a nonlinear relationship can give *r* near zero (6.3, nonlinear relationships).

2. Book questions and invented data

a. (example answer; many answers are possible) Third variables: parenting (for example, parents who use candy as a reward and set few limits), low self-control or impulsivity, or family stress could lead to both daily candy and later violence. Headline: "Children Who Eat Candy Daily More Likely to Be Arrested for Violence Later in Life". b. 1. Positive: students who sleep more tend to recall more words.

2. Large (above the ±.50 guideline) and positive.
3. Restriction of range: in this narrow range of sleep, the relationship disappears. Recruit students with a wide range of sleep habits, inspect the data and interpret *r* in light of the range.
4. No. Directionality: students with better memories may finish studying sooner and go to bed earlier. Third variable: stress (or caffeine, or workload) could reduce both sleep and recall. Only an experiment can show causation.
5. A nonlinear (upside-down U) relationship, like sleep and depression in the book; Pearson's *r* measures only straight-line relationships, so it would underestimate the relationship.

c. (example answers) (1) Unlikely: a long life cannot make people eat more lobster; third variable: wealth. (2) Present: weighing less may make exercise easier; third variable: age or general health. (3) Present: poor grades may lead students to drink more; third variable: low conscientiousness or a busy social life.

3. Language exercise

1. R. 2. C (*has a negative effect on*), although hedged with *perhaps*. 3. R.
4. (example answer) Students who slept more tended to recall more words; sleep duration was positively correlated with recall.
5. (example answer) Children who ate candy every day were more likely to be arrested for a violent offense later in life.

My notes